Name: Toby Huynh (thh45)

Date of submission: 02/23/2026

Problem set number: #3

Due date: 02/24/2026

Acknowledgement:
Fundamentals of Heat and Mass Transfer 8th Edition
Course notes

1) Known:

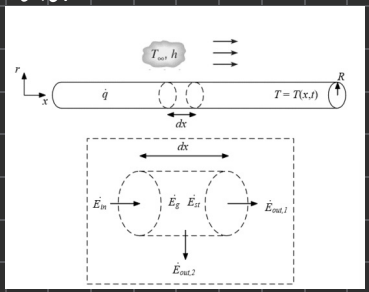
Radius: R

Volumetric heat gen. rate: $\dot{q}$

Ambient temp: T_∞

Heat transfer coeff.: h

Schematic:



Assumption:

- Uniform heat generation
- Small R
- Uniform temp. distribution in radial direction
- Constant k, ρ, c_p

Analysis:

a) $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2} - \dot{E}_{st} = 0 \rightarrow \dot{E}_{in} + \dot{E}_g = \dot{E}_{out,1} + \dot{E}_{out,2} + \dot{E}_{st}$

$$-k\pi R^2 \frac{\partial T}{\partial x}\Big|_x + \dot{q}(\pi R^2 dx) = -k\pi R^2 \frac{\partial T}{\partial x}\Big|_{x+dx} + h(2\pi R \cdot dx)(T - T_\infty) + \rho c_p (\pi R^2 dx)\frac{\partial T}{\partial x}$$

b) $$-k\pi R^2 \frac{\partial T}{\partial x}\Big|_x + k\pi R^2 \frac{\partial T}{\partial x}\Big|_{x+dx} = k\pi R^2 \frac{\partial^2 T}{\partial x^2} dx$$

Divide out $\pi R^2 dx$

$$k\frac{\partial^2 T}{\partial x^2} - \frac{2h}{R}(T - T_\infty) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

c) $T(0,t) = T_1$

At $x = L$: $-k\frac{\partial T}{\partial x}\Big|_{x=L} = h(T(L,t) - T_\infty)$

$T(x,0) = T_i$

2) Known:

$L_A = 0.30\,m$

$L_B = 0.10\,m$

$k_B = 30\,\frac{W}{m\,K}$

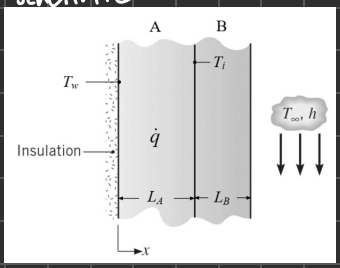
$h = 500\,\frac{W}{m^2 K}$

$T_\infty = 20°C$

$T_w = 115°C$

$T_i = 100°C$

Schematic:



Assumptions:

- Left of wall A is perfectly insulated
- No thermal resistance b/w A and B.

Analysis:

a) $q'' = h(T_s - T_\infty)$

$q'' = \frac{k_B(T_i - T_s)}{L_B}$

$\frac{k_B(T_i - T_s)}{L_B} = h(T_s - T_\infty)$

$\frac{30(100 - T_s)}{10 \times 10^{-2}} = 500(T_s - 20)$

$\frac{3000}{10 \times 10^{-2}} - 300T_s = 500T_s - 10000$

$40000 = 800\,T_s$

$T_s = 50°C = \boxed{323.15\,K}$

b) $q'' = \dot{q}\,L_A$

$\frac{k_B(T_i - T_s)}{L_B} = \frac{30(100-50)}{10 \times 10^{-2}} = 15{,}000\ W/m^2$

$\dot{q} = \frac{15000\,\frac{W}{m^2}}{30 \times 10^{-2}\,m} = \boxed{50000\,\frac{W}{m^3}}$

c) $k_A \frac{d^2T}{dx^2} + \dot{q} = 0 \quad (3.44)$

$\left.\frac{dT}{dx}\right|_{x=0} = 0$ since left wall is insulated

$T(L_A) = T_i = 100°C$

$T(0,0) = T_w = 115°C \longrightarrow C_2 = 115°C$

$\frac{d^2T}{dx^2} = -\frac{\dot{q}}{k_A}$

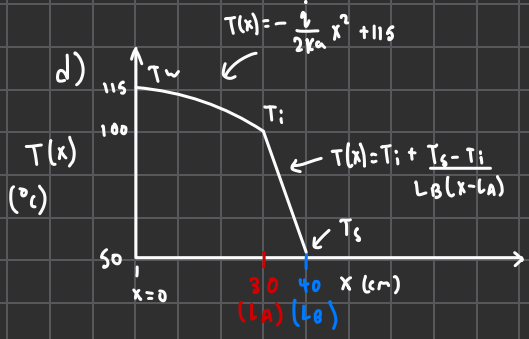
$\frac{dT}{dx} = -\frac{\dot{q}}{k_A}x + C_1 \longrightarrow \left.\frac{dT}{dx}\right|_{x=0} = 0 \longrightarrow C_1 = 0$

$T(x) = -\frac{1}{2}\frac{\dot{q}}{k_A}x^2 + C_1 x + C_2$

Thus, $T(x) = -\frac{1}{2}\frac{\dot{q}}{k_A}x^2 + 115°C \xrightarrow{x = L_A} \overset{T_i}{T(L_A)} = \frac{-\dot{q}L_A^2}{2k_A} + 115$

$k_A = \frac{\dot{q}L_A^2}{2(T_i - 115)} = \frac{-(50000\,\frac{W}{m^3})(30 \times 10^{-2}\,m)^2}{2(100°C - 115°C)} = \boxed{150\,\frac{W}{mK}}$

d)



3)

Known:

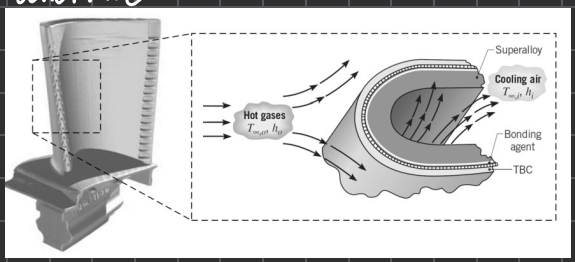
$k_b = 25 \frac{W}{mK}$, $k_{TBC} = 1 \frac{W}{mK}$, $R'' = 10^{-4} \frac{m^2 K}{W}$, $T_{\infty,o} = 1616K$, $T_{\infty,i} = 400K$, $h_o = 1000 \frac{W}{m^2 K}$, $h_i = 500 \frac{W}{m^2 K}$

$L_b = 5 \times 10^{-3} m$, $T_{max} = 1200°C$

Assumptions

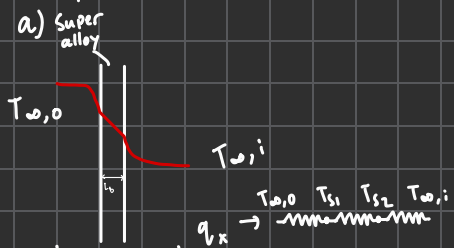
- metallic bonding agent has negligible thickness
- Blades at steady state condition, neglect radiation
- Blades are a plane wall, 1-D heat conduction through TBC and superalloy.

Schematic



Analysis:

a)



$$\frac{1}{h_o A} + \frac{L_B}{k_b A} + \frac{1}{h_i A} = R_{tot} \rightarrow R_{tot} A = R''_{tot}$$

$$R''_{tot} = \frac{1}{h_o} + \frac{L_B}{k_b} + \frac{1}{h_i}$$

$$R''_{tot} = \frac{1}{1000 \frac{W}{m^2 K}} + \frac{5 \times 10^{-3} m}{25 \frac{W}{mK}} + \frac{1}{500 \frac{W}{m^2 K}}$$

$$R''_{tot} = 0.0032 \frac{m^2 K}{W}$$

$$q'' = (T_{\infty,o} - T_{\infty,i}) / R''_{tot}$$

$$q'' = 380\,000 \frac{W}{m^2}$$

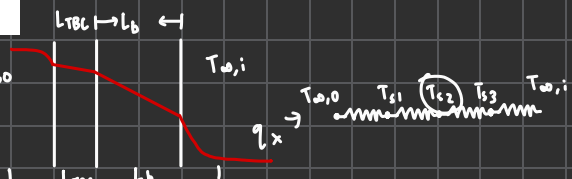
$$R'' = \frac{T_{\infty,o} - T_s}{q''} = \frac{1}{h_o}$$

$$T_s = T_{\infty,o} - \frac{q''}{h_o}$$

$$T_s = 1616K - \frac{380000 \frac{W}{m^2}}{1000 \frac{W}{m^2 K}}$$

$$\boxed{T_s = 1236K}$$ it can not be maintained below 1200K

b)



$$\frac{1}{h_o}, \frac{L_{TBC}}{k_{TBC}}, \frac{L_b}{k_b}, \frac{1}{h_i}$$

$$R''_{tot} = \frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + \frac{L_b}{k_b} + \frac{1}{h_i}$$

$$R''_{tot} = \frac{1}{1000 \frac{W}{m^2 K}} + \frac{0.5 \times 10^{-3} m}{1 W/mK} + \frac{5 \times 10^{-3} m}{25 W/mK} + \frac{1}{500 W/mK}$$

$$R''_{tot} = 0.0037 \frac{m^2 K}{W} + \underbrace{10^{-4} \frac{m^2 K}{W}}_{\text{Bonding agent}}$$

$$q'' = \frac{T_o - T_i}{0.0038 \frac{m^2 K}{W}} = \frac{1216K}{0.0038 \frac{m^2 K}{W}} \quad (3.14)$$

$$q'' = 320000 \frac{W}{m^2}$$

$$q'' = \frac{T_{\infty,o} - T_{s,1}}{1/h_o} \quad (3.16)$$

$$T_{s,1} = 1616K - \frac{320000 \frac{W}{m^2}}{1000 W/m^2 K} = 1296K$$

$$T_{s,2} = T_{s,1} - q''\left(\frac{L_{TBC}}{k_{TBC}}\right)$$

$$T_{s,2} = 1296K - \frac{(320000 \frac{W}{m^2})(0.5 \times 10^{-3})}{1 W/mK}$$

$$\boxed{T_{s,2} = 1136K}$$ yes, T_{max} can be maintained under 1200K